

DATA SCIENCE & AI LAB · MAY 2026 · GROUP 1

# Multimodal Damage Assessment for Insurance Claims

MILESTONE 2: DATASET PREPARATION

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# From Milestone 1 to a Training-Ready Dataset

Milestone 1 defined a multi-agent RAG system that reads damage photos and policy PDFs, detects and classifies damage, retrieves relevant policy clauses, and drafts a preliminary claim report with a LangGraph orchestrator routing each claim across four specialist agents and escalating low-confidence cases to human review.

## MILESTONE 2 OBJECTIVES

- 1 Dataset verification & download**  
Identify provenance, confirm licensing, download every dataset each agent needs.
- 2 Vision data preparation**  
Clean, remap, dedupe, augment and split VehiDE into training-ready artefacts.
- 3 Policy corpus construction**  
Author, chunk, embed and index a synthetic policy corpus for the Policy Agent.

## FOUR-AGENT ORCHESTRATION

### Damage Agent

YOLO detection of 6 damage classes

### Severity Agent

Bounding-box area-ratio scoring

### Policy Agent

RAG over synthetic policy corpus (MCP tool)

### Report Agent

LLM-based preliminary claim report

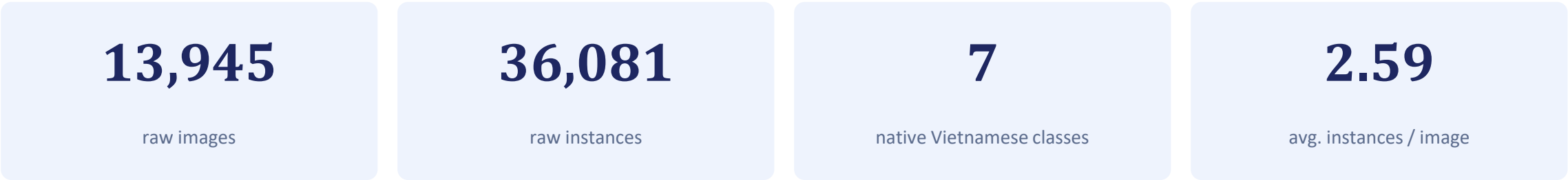
*Low-confidence outputs escalate to a human review queue.*

# Datasets

| Dataset                         | Feeds             | Role                                            | Status |
|---------------------------------|-------------------|-------------------------------------------------|--------|
| VehiDE (primary)                | Damage + Severity | 6-class YOLO fine-tuning & severity calibration | Ready  |
| Synthetic policy PDFs           | Policy Agent      | RAG retrieval corpus                            | Ready  |
| Synthetic incident descriptions | Report + Eval     | Ground-truth incident/report pairs              | Ready  |

*VehiDE is the largest publicly available annotated vehicle-damage dataset (13,945 images, 36,081 raw instances), selected as the primary training source. CarDD and the Severity dataset will be integrated once the training on model performs poorly.*

# VehiDE: Structure & Scale



## ANNOTATION SCHEMA (VIA-format JSON)

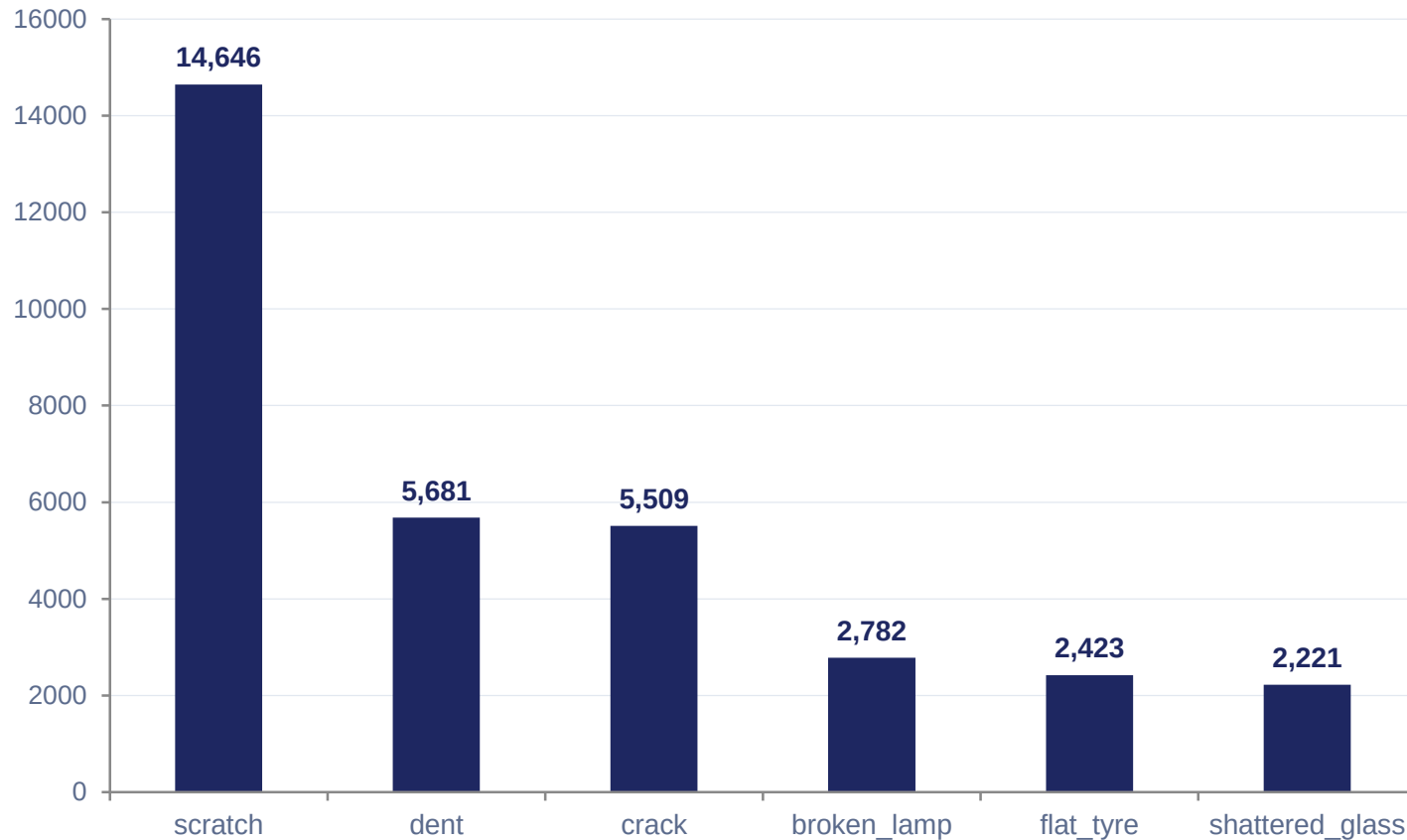
```
{
  "<file>.jpg": {
    "regions": [
      {"all_x": [...], "all_y": [...],
       "class": "rach"}, ...
    ]
  }
}
```

Bounding boxes are derived from each polygon's [x, y, width, height] extent.

## VIETNAMESE → PROJECT CLASS MAPPING

| Native                     | Project class   |
|----------------------------|-----------------|
| tray_son (paint_scratches) | scratch         |
| mop_lom (dents)            | dent            |
| rach (torn_body)           | crack           |
| be_den (broken_lamp)       | broken_lamp     |
| thung (puncture)           | flat_tyre       |
| vo_kinh (broken_glass)     | shattered_glass |
| mat_bo_phan (lost_parts)   | EXCLUDED        |

# Class Distribution & Imbalance

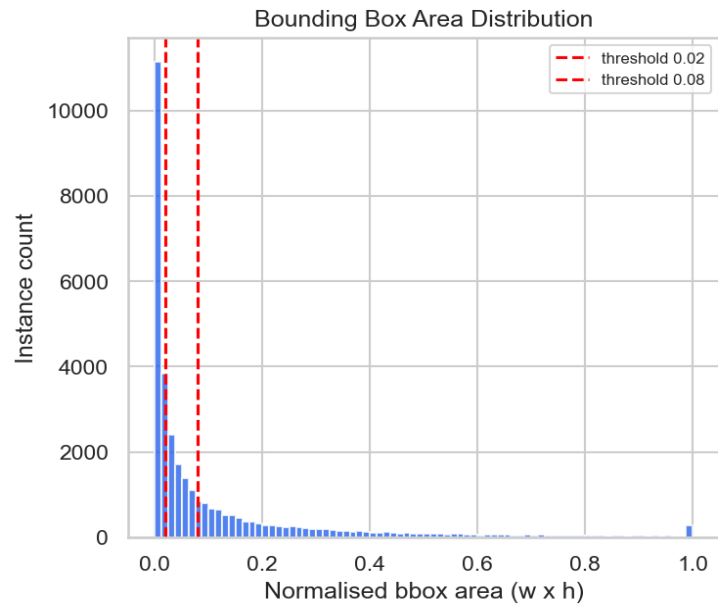


6.59 : 1

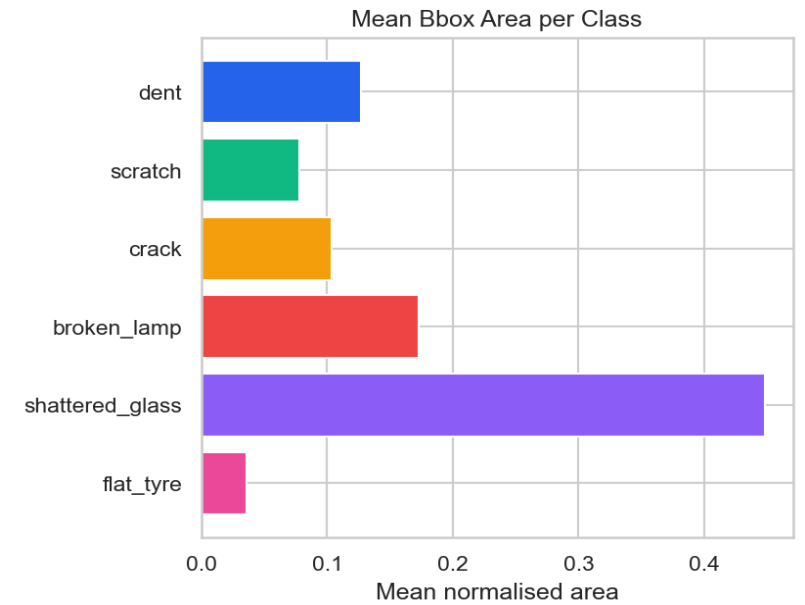
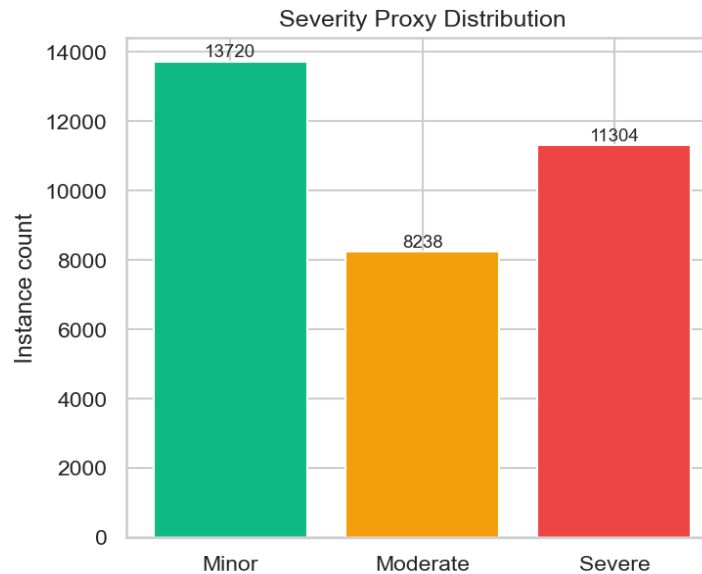
imbalance ratio · scratch vs. shattered\_glass

- scratch dominates at 44.0% of retained instances
- shattered\_glass is rarest at 6.7%
- torn\_body is termed as crack class
- Mitigated via class-weighted loss & oversampling of minority classes

# Bounding-Box Area & Severity Proxy



VehiDE — Bounding Box Area & Severity Proxy



Minor

13,720

41.3%

Moderate

8,238

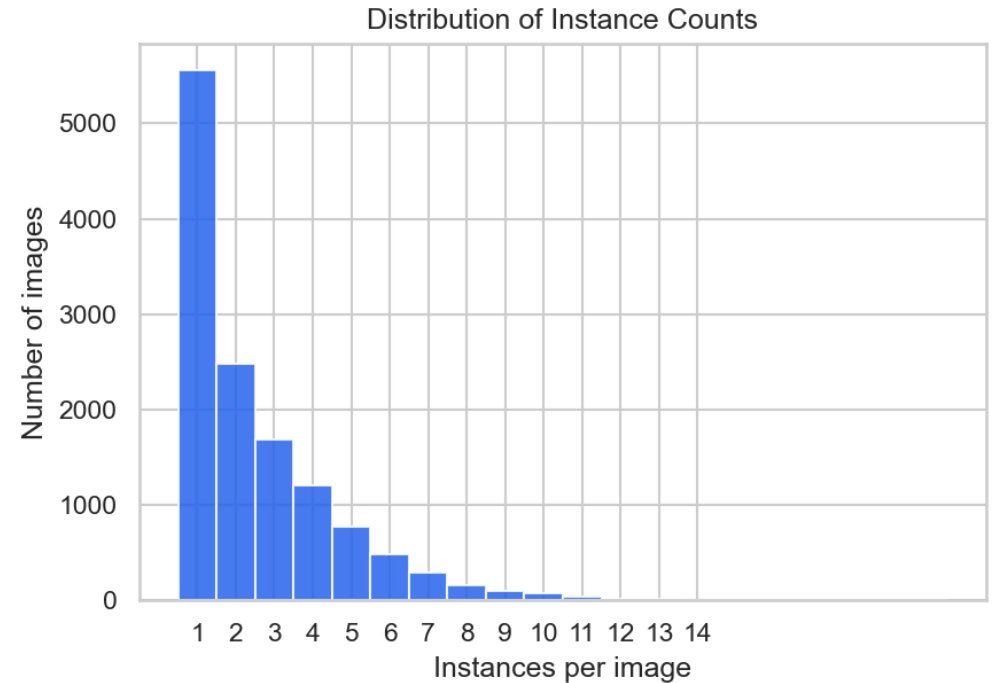
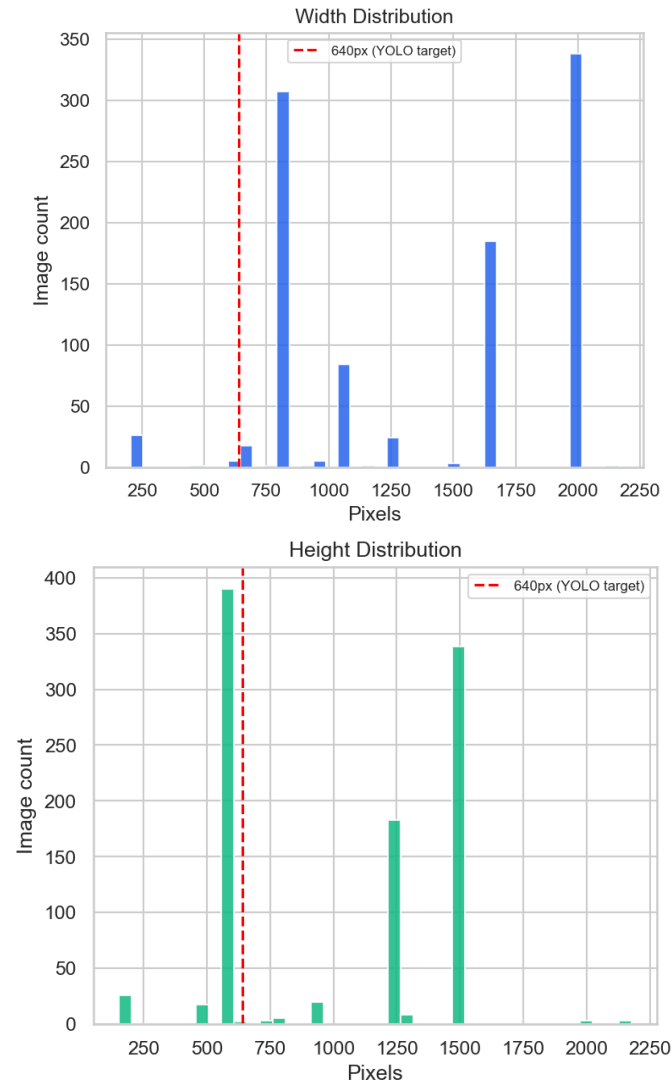
24.8%

Severe

11,304

34.0%

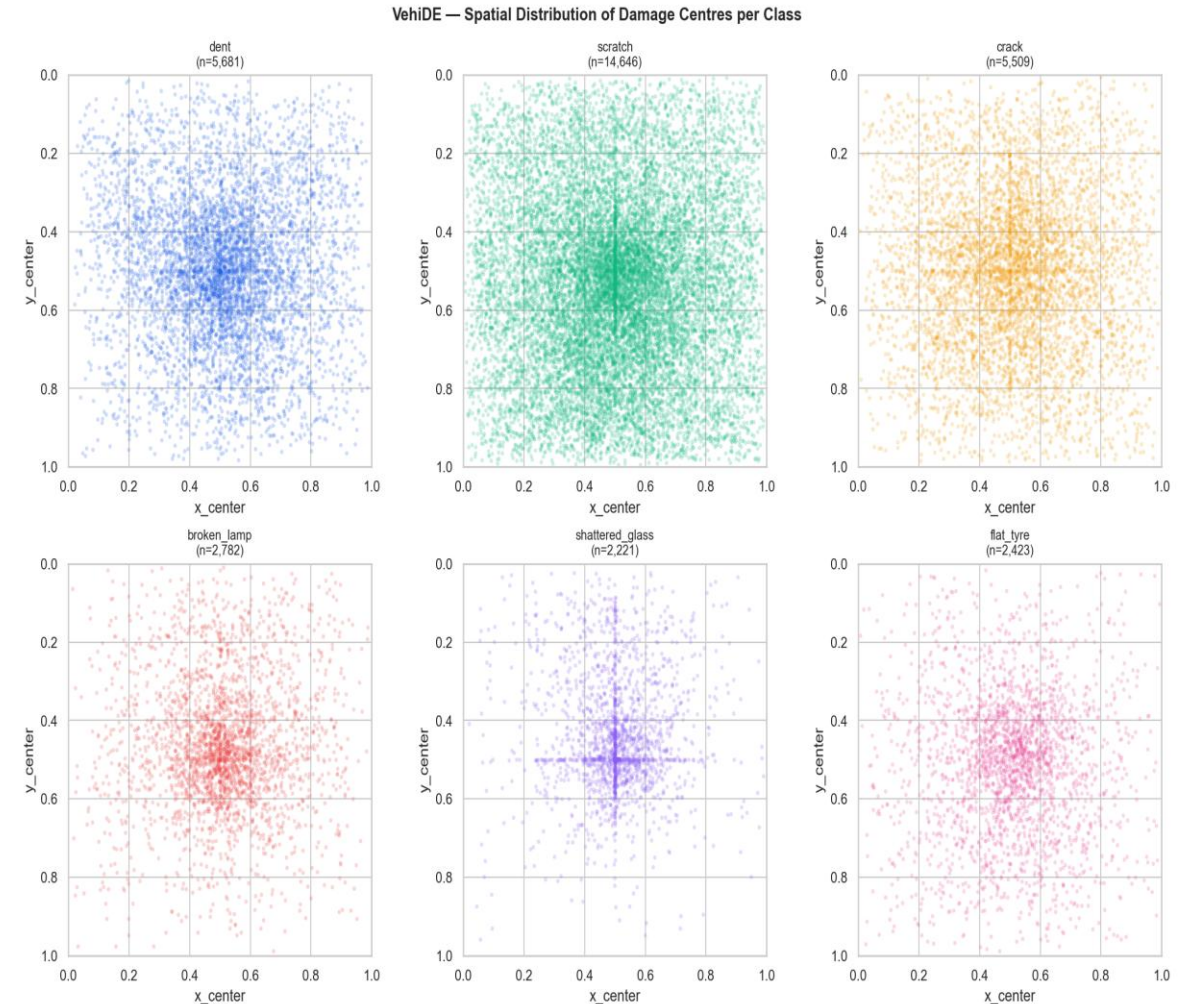
# Multi-Instance Images & Resolution Spread



Mean 2.58 / median 2.0 instances per image (max 20) multi-label detection is the norm.

Resolutions span 204–2,164 pixels wide.

# Where and With What Damage Appears





# Licensing, Privacy, Ethics & Reproducibility

## Source & Licensing

- VehiDE is distributed under the Apache License 2.0
- IRDAI policy PDFs used only as structural reference.

## Privacy

- Face / license-plate detector run across all 13,655 images, 0 flagged. Detector stays in the pipeline for future additions.

## Data Quality

- 0 corrupt files, 0 orphan images/annotations, 1 malformed region corrected
- 18 exact & 272 near-duplicates removed.

## Ethics & Bias

- VehiDE is Southeast-Asia-sourced, Indian vehicle types & conditions may be underrepresented.
- Mitigated via augmentation & flagged as a domain-shift risk.

# From Raw Download to Training-Ready Images

1

## Corrupt-file check

0 corrupt images found

2

## Class remap (7→6)

torn\_body → crack  
lost\_parts excluded

3

## YOLO format verify

4

## Letterbox resize

1280×1280, grey pad (114,114,114)

5

## Annotation spot-check

Random samples rendered & visually  
verified

6

## PII detection

Face/plate detector 0 flagged

7

## Deduplication

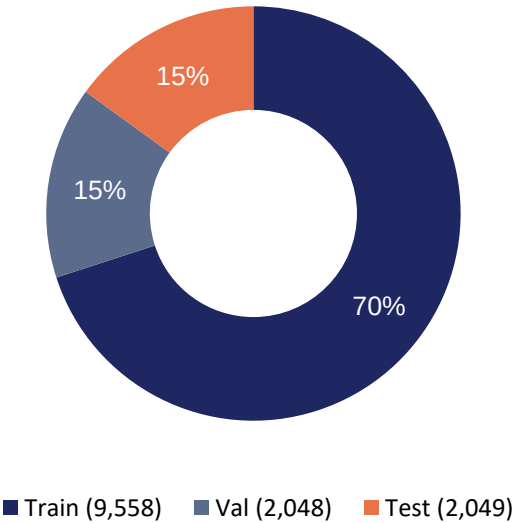
18 exact & 272 near-duplicates removed

8

## Stratified split

70 / 15 / 15 → train / val / test

# Balanced, Leakage-Checked, Training-Ready



13,655 images · 32,672 instances  
70 / 15 / 15 stratified split

## AUGMENTATION (TRAIN-TIME ONLY)

- Horizontal flip (p=0.5)
- Mosaic (4-image combine)
- HSV brightness / saturation jitter
- Small rotation & translation
- Motion blur mobile camera shake
- JPEG re-compression (quality 75)

## CLASS-TARGETED WEIGHTS (cls\_pw)

| Class           | Weight |
|-----------------|--------|
| scratch         | 1.0    |
| dent            | 2.6    |
| crack           | 2.7    |
| broken_lamp     | 5.3    |
| flat_tyre       | 6.0    |
| shattered_glass | 6.6    |

# RAG Corpus for the Policy Agent

5

synthetic PDFs

185

indexed chunks

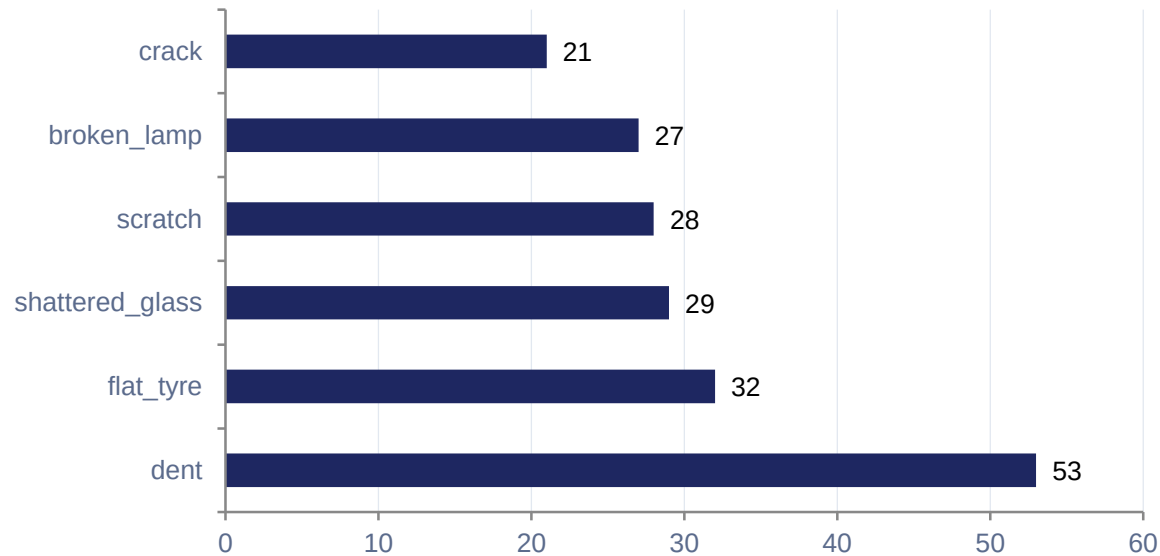
300 / 40

chunk size / overlap (tokens)

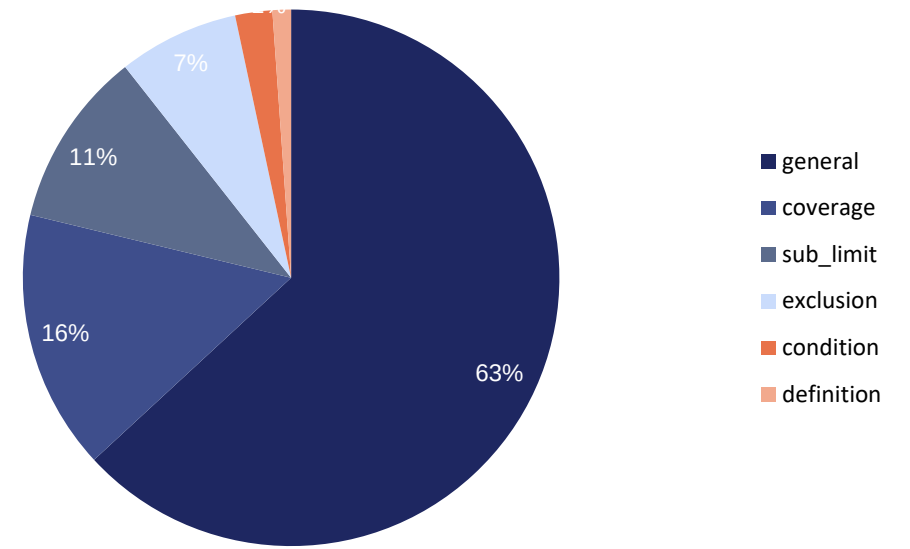
6

damage classes tagged

## CHUNKS PER DAMAGE CLASS



## CLAUSE TYPE DISTRIBUTION



# Ready for Model Training

13,655

images

32,672

retained instances

6

damage classes

185

policy chunks indexed

The data is ready for training immediately:

- Train / val / test splits finalised, leakage-checked, and logged as file lists.
- ChromaDB policy index (collection policy\_clauses, 185 chunks) built and query able.

# What's Ready, What's Next

## DELIVERABLES PRODUCED

- VehiDE EDA notebook + preprocessing script
- Cleaned & split image / label sets (train/val/test)
- Class remap lookup & YOLO config (damage.yaml, nc=6)
- Synthetic policy corpus + ChromaDB vector index (185 chunks)
- COCO Car Damage EDA (complete, used for comparison)
- Milestone 2 report

## PLANNED FOR MILESTONE 3

- Select model architecture: YOLO11m-seg vs. YOLOv8m-seg baseline
- Define LangGraph orchestrator state schema & MCP tool contracts
- Run YOLO baseline and establish mAP@50 & per-class F1
- Wire up Policy Agent's FastMCP retrieval tool; run precision check
- Validate Report Agent prompt template on 5 sample incident/image pairs
- And if the model performs poorly, then integrate CarDD data into the training dataset.

# Thank You

Questions & Discussion

Github: [HiveCase/Group-1-DS-and-AI-Lab-Project](#)  
Course: DS & AI Lab · May 2026 · Group 1

